SQL

GENERAL ASSEMBLY

Agenda

- What is SQL?
- SQL: The Software
- SQL: The Language
- BigQuery!

 © 2020 General Assembly

Learning Objectives

- Describe relational databases
- Why SQL?
- Implement many SQL commands over 3 lessons



What is SQL?



What is SQL?

Structured Query Language (SQL) is:

- Pronounced either “sequel” or “ess-kyu-ell”
- Is both the name of a language and a database software...

 © 2020 General Assembly

SQL The Database Software

A **SQL server** is a server that accepts **SQL queries** and stores data in a **SQL database**.

A SQL database is made up of many **SQL tables** (ie, dataframes).

But we already know how to store data in files! What's the advantage of having data stored in a SQL database.

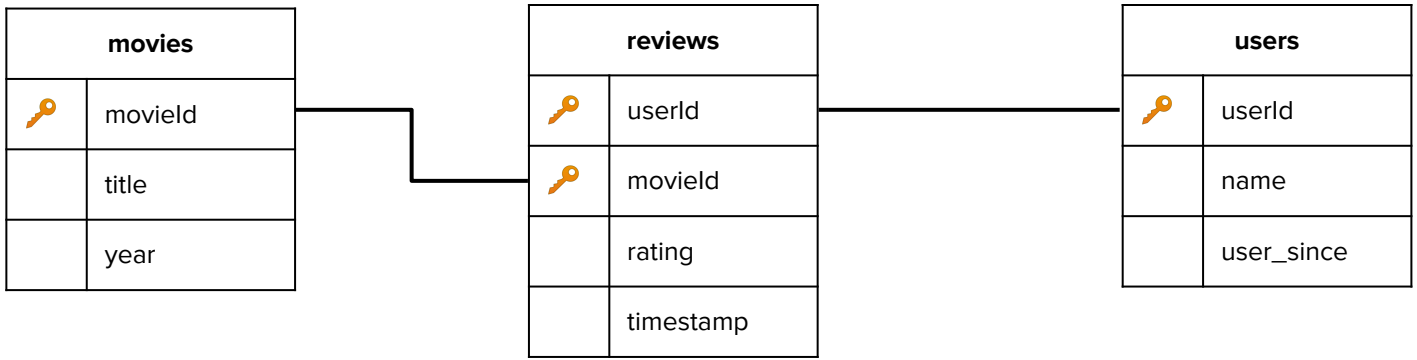
SQL The Database Software

A SQL database is...

- **Relational**
  - If set up properly, you can organize your tables to relate to one another via **keys**. Tables are **joined** on such keys so that you can query exactly the data you need easily from many tables.
- Secure (as in, “idiot-proof”)
  - **DBAs (database administrators)** can set up permissions structures in SQL so that the intern doesn’t accidentally delete important tables

SQL ERDs

Again, if things are set up properly, an **entity relationship diagram (ERD)** can be made (or generated!). This is a graph that describes the relationships between tables in your SQL DB. For example:



 © 2020 General Assembly

SQL Join Example

userId	movieId	rating	timestamp
1	1	5	1/1/22 1:00
2	1	4	1/1/22 2:00
2	2	2	1/2/22 11:00

userId	name	user_since
1	Tim Book	2/28/2019
2	Katie Sylvia	8/17/2020

movieId	title	year
1	Shrek	2001
2	The Hulk	2008

userId	movieId	rating	timestamp	name	user_since	title	year
1	1	5	1/1/22 1:00	Tim Book	2/28/2019	Shrek	2001
2	1	4	1/1/22 2:00	Tim Book	2/28/2019	Shrek	2001
2	2	2	1/2/22 11:00	Katie Sylvia	8/17/2020	The Hulk	2008

SQL The Database Software

A SQL database is...

- Redundant (as in “safe”)
  - DBAs can set SQL DBs to be periodically backed-up in case something goes wrong
  - DBs are often stored in redundant hardware on cloud servers so that if one goes down or is destroyed, data aren’t lost.

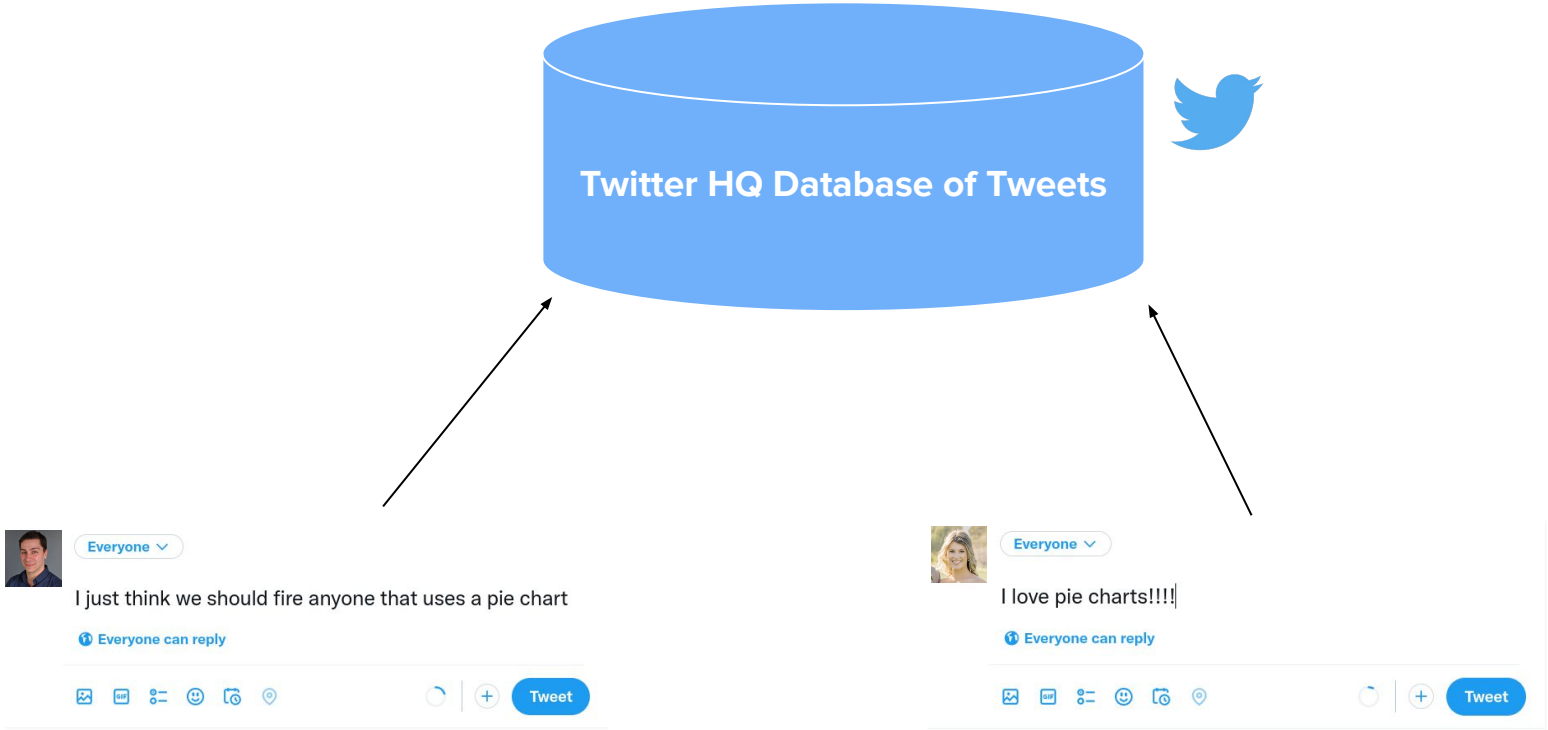
SQL The Database Software

A SQL database is...

- Parallel
  - SQL DBs are designed so that multiple users and edit and query simultaneously.
  - SQL can also **lock** rows in a table when one person is editing it so another person doesn't also try to edit i
  - DBs are often large in terms of storage and powerful in terms of CPU strength. Depending on the architecture, SQL can handle large, distributed calculations!

 © 2020 General Assembly

Parallel SQL Operations



The SQL Language

SQL is also the programming language that is used to query a SQL database!

SQL is...

- It is a **domain-specific, declarative** language
  - domain-specific = only good for one domain (data stuff)
  - declarative = specify only *what* is to be done, not *how*
- As far as languages go, it is very old (1960s!)
- Very easy to pick up. You will be able to handle 99% of queries after our 3 lessons

The SQL Language

SQL is also the programming language that is used to query a SQL database!

SQL is...

- Basically dead as a **standalone language**
- Industry-standard and thriving as a language invoked via **glue languages** such as Python, R, JavaScript, Ruby, etc

The SQL Language

SQL is also the programming language that is used to query a SQL database!

SQL is...

- **Not just one language!**
- While there is a “SQL standard”, different versions of SQL do things pretty differently sometimes.
- These differences aren’t worth memorizing... but they’re worth Googling when you encounter something that should work but doesn’t!

 © 2020 General Assembly

Some SQL Dialects

There are many “dialects” of SQL. Here are just a few:

- **Postgres**: Open source, probably the most common
- **MySQL**: Also open source, but distributed by Oracle
- **Transact-SQL (T-SQL)**: SQL built by Microsoft for MS SQL Server
- **SQLite3**: A light and portable version of SQL often used to build small examples (often in Python). Contained in the Python stdlib!
- **HiveQL (HQL)**: Ok technically not a SQL dialect, but it looks pretty identical to SQL and actually powers the backend of **Spark**!

Google BigQuery

Google BigQuery is a dialect of SQL for use in GCP. It specializes in handling large, distributed datasets at scale. More importantly for us today, Google provides many example databases for us to use for free!



Google Big Query

 © 2020 General Assembly

Let's Do It!




GA